

Comparative Evaluation of Data Sanitization Strategies for Applied Behavior Analysis and Intellectual/Developmental Disabilities Documentation: Prioritizing Safety Through Multi-System Care Coordination and Predictive Analytics with Reduced PHI Exposure

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Executive Summary

Healthcare organizations serving individuals with intellectual and developmental disabilities (IDD) face a critical challenge: deploying AI-powered clinical decision support systems while protecting the privacy of vulnerable populations under multiple regulatory frameworks (HIPAA, FERPA, 42 CFR Part 2, OPWDD). Current approaches to data sanitization either strip too much semantic content needed for clinical utility or leave residual re-identification risk unacceptable for small IDD populations where behavioral details and rare condition combinations create unique identifying profiles. No systematic research exists comparing sanitization strategies specifically for Applied Behavior Analysis (ABA) documentation contexts, particularly for the most challenging use cases: multi-system care coordination and predictive analytics.

This research represents the first systematic evaluation of data sanitization specifically tailored to IDD/ABA service contexts, filling a critical gap in privacy-preserving AI literature while delivering immediately actionable recommendations for healthcare organizations serving vulnerable populations. By combining rigorous quantitative evaluation with practitioner-validated usability assessment, this work bridges the gap between privacy research and real-world implementation constraints, enabling safer AI adoption in high-risk healthcare domains.

Short Biography

Graduate student specializing in machine learning, deep learning, and data analytics with focused expertise in medical and clinical applications. Current research centers on privacy-preserving AI for vulnerable populations, particularly individuals with intellectual and developmental disabilities (IDD). Academic background includes coursework in advanced neural network architectures, statistical analysis, bioinformatics, and healthcare technology. Practical experience encompasses multiple machine learning projects using Python and R, Kaggle competition participation, and medical imaging/clinical data analysis. Research interests lie at the intersection of clinical AI applications, privacy-preserving techniques, and ethical deployment of machine learning systems in healthcare settings serving vulnerable populations. This capstone project represents the synthesis of technical AI/ML expertise with domain knowledge in Applied Behavior Analysis (ABA) documentation and regulatory frameworks governing IDD services.

1 Problem statement & Key literature

Healthcare organizations increasingly want to use LLM-powered AI for tasks such as clinical documentation assistance, multi-system care coordination, and predictive analytics, but adoption is constrained by patient privacy concerns, complex regulatory requirements, and the absence of clear, evidence-based guidance on data sanitization strategies (Kaissis et al., 2022; U.S. Department of Health & Human Services [HHS], 2012). Narrative clinical text often contains rich combinations of direct identifiers (e.g., names, dates of birth, addresses, medical record numbers), quasi-identifiers (e.g., rare conditions, family structures, service locations), and sensitive behavioral details that can re-identify individuals even after obvious identifiers are removed (El Emam et al., 2009; Sweeney, 2002). Existing guidance such as HIPAA Safe Harbor focuses on removing 18 categories of identifiers, yet multiple studies and regulatory commentaries highlight that this is often insufficient for free-text clinical notes and high-risk, small-population domains like behavioral health and intellectual/developmental disabilities (El Emam et al., 2009; HHS, 2012).

At the same time, the healthcare AI literature describes a range of technical approaches—traditional de-identification pipelines, replacement and masking strategies, hashing and pseudonymization, and synthetic or partially synthetic text generation—but there is limited comparative, use-case-oriented evidence on how these strategies perform when evaluated simultaneously on (a) PHI exposure reduction, (b) preservation of downstream AI utility, and (c) production deployment scalability (Kaissis et al., 2022). There is even less guidance tailored to multi-system care coordination (e.g., linking data across disability services, behavioral health, educational, and medical systems) and predictive analytics, which are among the most sensitive and challenging use cases identified by partners such as Upstate Caring Partners (J. Jaramillo, personal communication, February 12, 2026).

This capstone addresses the following core problem: Healthcare organizations lack a rigorous, empirically validated framework for selecting and implementing data sanitization strategies that enable LLM-powered AI applications while prioritizing maximal PHI exposure reduction for patients and persons we serve. The project aims to fill this gap by developing and evaluating a multi-strategy sanitization framework—covering replacement, masking, hashing, and redaction—on synthetic patient data, and by producing actionable recommendations, metrics, and implementation artifacts that can be used by healthcare and human services organizations to safely adopt AI technologies (Kaissis et al., 2022; Sweeney, 2002).

2. Solution approach to the problem

This capstone develops and evaluates a multi-strategy data sanitization framework that sits upstream of LLM-powered AI applications and focuses on safety-first PHI reduction while still considering utility and deployment feasibility. The approach is shaped directly by UCP's priorities.

2.1 Custom PHI Recognizers for High-Risk Identifiers

The framework begins with development and configuration of PHI recognizers tailored to high-risk healthcare and human services identifiers. At minimum, these include:

- First name and last name
- Date of birth
- Residential addresses (street, city, state, zip), explicitly excluding site names where required
- Medicaid number in an 8-character alphanumeric format
- Additional common healthcare identifiers (e.g., medical record numbers, phone numbers, emails) as needed for generalizability

These recognizers will combine pattern-based rules and NLP models where appropriate, with an emphasis on high recall to avoid missed identifiers.

2.2 Four Sanitization Strategies

On top of the recognizers, the framework implements four sanitization strategies:

- **Replacement:** Replace detected identifiers with descriptive placeholders or pseudonyms (e.g., “[PATIENT_NAME]”, “[DATE]”), preserving sentence structure and context while removing PHI.
- **Masking:** Partially obscure identifiers (e.g., truncating or masking portions of an ID or date) to reduce identifiability while retaining some structural information useful for quality checks or internal workflows.
- **Hashing:** Apply deterministic cryptographic hashing to identifiers (e.g., salted hashes) to support re-linkage across records or systems without revealing the raw identifier values.

- **Redaction:** Completely remove identifiers or replace them with generic tokens when safety requires maximal suppression, especially in high-risk scenarios.

2.3 Evaluation on Synthetic Healthcare Data and UCP-like Use Cases

The framework will be tested on approximately 5,000 synthetic patient records that simulate realistic clinical narratives and identifier patterns. The evaluation will explicitly incorporate two UCP-prioritized use cases:

- **Multi-System Care Coordination:** Simulated record linkage across synthetic “systems” (e.g., disability services, behavioral health, education, medical) to evaluate whether sanitized data still supports accurate coordination while meeting PHI reduction thresholds.
- **Predictive Analytics:** Training and evaluating simple predictive models (e.g., classification or regression) using original versus sanitized data to quantify the impact of each strategy on model performance under strict PHI reduction constraints.

2.4 Safety-First Evaluation Framework and Implementation Artifacts

The evaluation hierarchy will be: (1) PHI exposure reduction for PWS, (2) use case enablement (coordination and predictive analytics), and (3) implementation feasibility and performance. This means that strategies failing to meet PHI reduction thresholds will not be considered acceptable, even if they preserve utility. The final deliverables will include:

- A sanitization strategy comparison and selection matrix
- A modular, API-based implementation that can be integrated upstream of LLM applications
- Guidance on how the sanitization layer interacts with data classification and access control policies in healthcare and human services organizations

This approach directly supports the partner’s priorities by rigorously focusing on sanitization as the primary data governance control, quantifying safety and utility for critical AI use cases, and delivering reusable technical assets.

3 Project Milestones

The project will be executed in three major milestones over a 15-week capstone period, with each milestone building upon the previous to deliver a comprehensive data sanitization evaluation framework.

3.1Timeline Overview

Milestone	Weeks 1-5	Weeks 6-10	Weeks 11-15	Duration	Effort	Status Check
Milestone 1: Data, Recognizers, Baseline				5 weeks	50-60 hrs	Week 5
Milestone 2: Sanitization Strategies				5 weeks	50-60 hrs	Week 10
Milestone 3: Use Cases & API				5 weeks	50-60 hrs	Week 15

3.2 Milestone Breakdown

MILESTONE 1: Data, Recognizers, and Baseline Setup (Weeks 1-5)	
Objectives	• Design synthetic data schema and generate ~110 synthetic ABA/IDD clinical records • Implement baseline PHI recognizers for UCP-specific identifiers (First Name, Last Name, DOB, Address, Medicaid ID) • Establish evaluation datasets with ground-truth PHI annotations
Key Deliverables	✓ 110 synthetic clinical records (40 behavior logs, 30 session notes, 20 progress reports, 10 BIPs, 10 IEPs) ✓ Custom Presidio recognizers with $\geq 95\%$ recall on validation set ✓ Ground truth annotation dataset (100-150 test cases)
Success Criteria	• Recognizer precision $\geq 90\%$, recall $\geq 95\%$ on held-out test set • UCP clinician validation: realism rating $\geq 4.0/5.0$ • Advisor approval to proceed to Milestone 2

MILESTONE 2: Implement and Evaluate Sanitization Strategies (Weeks 6-10)
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Objectives	• Implement two sanitization strategies (semantic redaction, cryptographic hashing) as modular components • Apply each strategy to full synthetic dataset and quantify PHI exposure reduction • Measure semantic utility using text similarity and clinical information retention
Key Deliverables	✓ Sanitized datasets for each strategy (220 total documents) ✓ Quantitative evaluation report (PHI reduction and utility metrics) ✓ UCP QA team Session 1 feedback incorporated
Success Criteria	• At least one strategy achieves $\geq 95\%$ PHI exposure reduction • Semantic similarity ≥ 0.75 for best-performing strategy • Statistical significance demonstrated (ANOVA, t-tests, effect sizes)

MILESTONE 3: Use Case Evaluation, Integration, and Guidance (Weeks 11-15)	
Objectives	• Evaluate impact on multi-system care coordination and predictive analytics use cases • Develop evidence-based recommendation matrix for UCP • Provide data classification and access control integration guidance
Key Deliverables	✓ Use case evaluation results (coordination linkage accuracy, predictive model performance) ✓ Strategy recommendation matrix with implementation guidance ✓ Complete research report (30-35 pages) with statistical analysis ✓ UCP QA team final approval (Session 3)
Success Criteria	• Multi-system coordination: linkage accuracy $\geq 98\%$ • Predictive analytics: model F1-score $\geq 75\%$ on sanitized data • UCP QA team approval $\geq 80\%$ for recommended strategies • Complete deliverables ready for UCP implementation

4 Proposed outcomes

The project is expected to produce the following outcomes:

- I. **A Multi-Strategy Sanitization Framework for Healthcare AI**
A structured framework that compares replacement, masking, hashing, and redaction strategies under a safety-first evaluation hierarchy, with explicit attention to multi-system coordination and predictive analytics.
- II. **Custom PHI Recognizers Covering UCP-Prioritized Identifiers**
Configured and evaluated PHI recognizers for first name, last name, DOB, residential addresses, and 8-character Medicaid IDs, along with other key healthcare identifiers, with documented performance on synthetic data.
- III. **Quantitative Evidence on Privacy–Utility Tradeoffs**
Empirical results demonstrating how each strategy performs on PHI exposure reduction, semantic and task-level utility, and deployment-related metrics, enabling reasoned tradeoff decisions.
- IV. **API Prototype and Implementation Assets**
A minimal but realistic API that can sanitize text in real time prior to LLM processing, along with code, configuration examples, and documentation suitable for adaptation by healthcare and human services organizations.
- V. **Evidence-Based Guidance for Practitioners**
A strategy selection matrix and integration guidance that help organizations choose and configure sanitization approaches that align with their risk tolerance, regulatory environment, and AI use cases, particularly when serving vulnerable populations.

5 Research Methodology

5.1 Research Design

Mixed-Methods Evaluation Study with quantitative comparative analysis as primary component and qualitative QA team validation as secondary component.

Study Type: Comparative evaluation research focused on evidence-based recommendations supported by rigorous empirical analysis, not production software engineering.

5.2 Data Source and Synthetic Document Generation

Real ABA/IDD documentation contains highly sensitive information unsuitable for research sharing or academic projects without extensive IRB approval and organizational data use agreements. Synthetic data enables rigorous testing without privacy violations while allowing controlled insertion of known PII for ground-truth evaluation.

Synthetic Document Generation Process:

1. Develop templates based on published ABA documentation examples (progress notes, functional behavior assessments, behavior intervention plans)
2. Generate 110 realistic synthetic documents using:
 - Name generation from public name frequency datasets (U.S. Census data)
 - Realistic demographic distributions matching IDD population characteristics
 - Behavioral descriptors and clinical narratives reflecting typical ABA documentation patterns
 - Controlled insertion of UCP-specific identifier formats (8-char Medicaid IDs, address formats, DOB patterns)

3. Realism validation: 2-3 experienced ABA clinicians rate synthetic documents on clinical authenticity (target: mean rating ≥ 4.0 on 5-point scale)

Dataset Characteristics:

- **Size:** 110 synthetic documents (sufficient for statistical power in paired comparisons)
- **Document types:** Progress notes (40%), Functional Behavior Assessments (30%), Behavior Intervention Plans (30%)
- **Identifier density:** Average 8-12 identifiable entities per document across five identifier types

5.3 Sanitization Strategies

Strategy 1: Semantic Redaction (Primary Focus – 80% Effort)

- Detect entities using custom Presidio recognizers
- Replace with semantic-preserving generic tokens:
 - Names → [CLIENT_NAME], [STAFF_NAME], [FAMILY_MEMBER]
 - DOB → [DATE_OF_BIRTH]
 - Addresses → [STREET_ADDRESS], [CITY], [STATE], [ZIP]
 - Medicaid ID → [MEDICAID_ID]
- Maintains document structure and readability while removing all raw PHI

Strategy 2: Cryptographic Hashing (Secondary Focus – 20% Effort)

- Detect entities using same custom recognizers
- Apply deterministic salted SHA-256 hashing: $\text{hash} = \text{SHA256}(\text{entity_value} + \text{salt})$
- Replace entities with truncated hash values (first 16 characters for readability)
- Enables cross-system linkage without exposing raw identifiers
- Salt management: secure storage, rotation policy considerations

5.4 Evaluation Metrics

Primary Safety Metrics

PHI Exposure Reduction Rate:

$$\text{Reduction Rate} = \frac{\text{Correctly Removed Entities}}{\text{Total True Entities in Document}} \times 100\%$$

Calculated per document, aggregated across dataset. Target: $\geq 95\%$.

Residual Re-identification Risk Score:

- Assess uniqueness of remaining quasi-identifier combinations (e.g., "17-year-old with autism diagnosis receiving services in rural upstate NY")
- Calculate k-anonymity: minimum group size sharing same quasi-identifier values

- Binary risk flag: "Could this document plausibly re-identify an individual?" (QA team judgment)

Entity Removal Completeness:

- Manual spot-checking (20% random sample, n=22 documents)
- Verification: Detected entities actually removed in sanitized output
- Formula: (Entities successfully removed) / (Total detected entities)

Secondary Use Case Metrics

Multi-System Care Coordination – Linkage Accuracy:

- Simulate two "systems" (split synthetic dataset in half)
- Use hashed identifiers as linking keys to match records across systems
- Calculate: (Correctly matched records) / (Total records) × 100%
- Target: ≥90% linkage accuracy with ≥97% PHI reduction

Predictive Analytics – ML Model Performance:

- **Task:** Binary classification predicting behavioral outcome (e.g., intervention success/failure) from synthetic document text
- **Baseline:** Random Forest classifier (100 trees, max_depth=10) trained on original synthetic documents
- **Treatment:** Same classifier architecture trained on sanitized documents (both strategies separately)
- **Metric:** Accuracy degradation = $|Acc_original - Acc_sanitized| / Acc_original \times 100\%$
- **Target:** <15% degradation with ≥95% PHI reduction

Tertiary Feasibility Metrics

QA Team Usability Survey (n=3-5):

- Likert scale (1-5): "Sanitized documents preserve sufficient information for care coordination decision-making"
- Open-ended: Identify adoption barriers, implementation concerns
- Clinical authenticity rating post-sanitization

Computational Cost:

- Processing time per document for each strategy
- Scalability assessment: projected time for 10,000-document corpus

5.5 Statistical Analysis

Comparative Analysis :

- **Paired t-test:** Compare PHI reduction rates between redaction and hashing strategies on same 110 documents
- **Effect size:** Calculate Cohen's d to quantify practical significance of differences
- **Confidence intervals:** 95% CI for mean PHI reduction rates per strategy

Subgroup Analysis:

- Stratify by document type (progress notes vs. FBAs vs. BIPs)
- Assess whether strategy performance varies by identifier density or document length

Validation:

- Inter-rater reliability (Krippendorff's alpha) for QA team residual risk assessments
- Sensitivity analysis: How do results change if precision/recall thresholds vary?

5.6 Quality Assurance and Validation

QA Team Composition: 3-5 participants including ABA clinicians (BCBAs preferred), data governance staff, and/or privacy compliance personnel.

Review Protocol:

1. Present 20 randomly sampled sanitized documents (10 per strategy)
2. Reviewers assess: (a) Residual re-identification risk, (b) Clinical usability, (c) Adoption feasibility
3. Semi-structured interviews (30-45 min each) to capture qualitative insights
4. Thematic analysis of interview transcripts identifying common concerns and recommendations

6 Limitations and Mitigation

Limitation 1: Synthetic Data Generalizability

Synthetic documents may not capture all patterns in real clinical practice, potentially affecting external validity.

Mitigation: Validate synthetic realism with experienced clinicians before analysis; clearly scope findings to synthetic data context; recommend future validation with real (properly de-identified) datasets in collaborative partnerships.

Limitation 2: Limited QA Sample Size

n=3-5 reviewers provides directional evidence but limited statistical power for usability conclusions.

Mitigation: Use mixed-methods triangulation (quantitative metrics + qualitative themes); characterize sample demographics; explicitly scope usability findings.

Limitation 3: Use Case Simulation

Multi-system coordination and predictive analytics tested via simulation, not real operational deployment.

Mitigation: Design realistic simulations grounded in UCP's actual use case requirements; recommend pilot deployment as Phase 2 future work.

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